

Jayanthkumar Karthik

+1 (520) 491-0014 | jayanthkumar.karthik13@gmail.com | [LinkedIn](#) | [GitHub](#)

Summary

Software Engineer with 4+ years of experience building scalable backend systems, microservices, and full-stack applications across banking, fintech, and enterprise domains. Proficient in Java, Spring Boot, Python, C++, React.js, Node.js, Kafka, and cloud technologies including AWS, Docker, and Kubernetes. Experienced in designing distributed systems, RESTful APIs, event-driven architectures, CI/CD pipelines, and performance optimization for high-volume applications. Skilled in AI-powered solutions using LLMs, LangChain, RAG, and OpenAI technologies, with a strong focus on delivering secure, reliable, and business-driven software solutions in Agile environments.

Skills

- Programming Languages:** Java, Python, C++, C#, JavaScript, TypeScript, SQL
- Backend Development:** Spring Boot, Node.js, Express.js, Microservices, RESTful APIs, Distributed Systems, Event-Driven Architecture, API Development
- Frontend Development:** React.js, Angular, Redux, JavaScript, TypeScript, Responsive Web Applications
- Cloud Technologies:** AWS, AWS Lambda, Amazon EC2, Amazon EKS, Amazon S3, Amazon RDS, Cloud-Native Applications, Serverless Architecture
- AI & Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), LangChain, OpenAI APIs, Generative AI, Prompt Engineering
- Messaging & Streaming:** Apache Kafka, Kafka Streams, Event-Driven Systems, Asynchronous Processing
- Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, SQL, Database Design, Query Optimization
- DevOps & CI/CD:** Docker, Kubernetes, Jenkins, Terraform, Git, GitHub Actions, CI/CD Pipelines, Infrastructure as Code (IaC)
- Monitoring & Observability:** Amazon CloudWatch, Prometheus, Grafana, ELK Stack, Logging, Monitoring, Alerting, Incident Management
- Software Engineering:** Object-Oriented Programming (OOP), System Design, Design Patterns, Secure Coding, Debugging, Troubleshooting, Unit Testing, SDLC, Agile Scrum
- Architecture & Integration:** Microservices Architecture, Distributed Computing, Third-Party API Integration, Payment Gateway Integration, Scalable Systems Design
- Professional Skills:** Cross-Functional Collaboration, Stakeholder Management, Requirement Analysis, Technical Documentation, Mentoring, Problem Solving, Team Collaboration

Experience

Software Engineer

Fidelity Investment, USA

Jan 2026 – Present

- Spearheaded development of distributed payment processing microservices using Java, Spring Boot, Kafka, and AWS Lambda, to modernize transaction orchestration workflows, reducing processing latency by 35% and improving payment reliability for digital banking customers.
- Engineered secure REST APIs and event-driven services using Spring Boot, Kafka, and Redis, to support real-time account servicing and fraud monitoring capabilities, enabling faster transaction validation for risk management and compliance teams.
- Designed and implemented AI-powered customer assistance solutions using Python, OpenAI APIs, LangChain, and RAG architectures, to automate policy and transaction inquiries, reducing manual support effort by 40% for operations teams.
- Modernized cloud-native applications using Docker, Kubernetes, AWS EKS, and Terraform, to improve deployment scalability and platform resilience, decreasing production incidents and lowering MTTR by 30% for engineering stakeholders.
- Developed React.js and TypeScript-based customer servicing dashboards, integrating backend APIs and analytics services, to provide real-time account insights and transaction visibility for customer service representatives.
- Optimized high-volume data pipelines using Kafka Streams, PostgreSQL, and distributed caching strategies, to process millions of financial events daily, improving system throughput and supporting enterprise-scale banking operations.
- Collaborated with architects, product owners, and security teams to implement observability frameworks using CloudWatch, Prometheus, Grafana, and ELK Stack, to strengthen monitoring coverage and improve operational decision-making across critical banking platforms.

Software Engineer

Affirm, India

Jun 2021 – Dec 2023

- Developed scalable microservices using Java, Spring Boot, and Kafka, to support enterprise digital transformation initiatives, improving transaction processing efficiency and enabling seamless integration across multiple business applications.
- Architected RESTful APIs and backend services using Spring Boot, Node.js, and Express.js, to streamline customer onboarding and workflow automation processes, reducing manual processing effort for business operations teams.
- Implemented cloud-native solutions on AWS utilizing EC2, S3, Lambda, and RDS services, to improve application scalability and operational flexibility, supporting high-availability enterprise platforms.
- Built responsive web applications using React.js, Angular, TypeScript, and JavaScript, to deliver intuitive user experiences and self-service capabilities, increasing user engagement and reducing support dependencies.
- Automated CI/CD pipelines using Jenkins, Docker, Kubernetes, and GitHub Actions, to accelerate software delivery cycles, reducing deployment effort by 50% and improving release consistency for development teams.
- Enhanced data processing workflows using Python, SQL, Kafka, and PostgreSQL, to support reporting and business intelligence requirements, improving data availability and decision-making for stakeholders.
- Partnered with cross-functional teams in Agile environments to modernize legacy applications into microservice-based architectures, improving maintainability, scalability, and long-term operational efficiency across enterprise systems.

Software Developer

Paytm, India

Mar 2020 – May 2021

- Developed backend services using Java, Spring Boot, and REST APIs, to support wallet, recharge, and payment workflows, enabling reliable transaction processing for millions of platform users.
- Implemented high-performance modules using C++ and multithreading techniques, to optimize transaction execution and memory utilization, improving processing efficiency by 20% for payment gateway services.
- Built event-driven integrations using Kafka and Spring Boot, to facilitate secure communication between payment, merchant, and settlement systems, improving transaction consistency across distributed services.
- Created responsive frontend features using React.js, JavaScript, and Redux, to enhance merchant and customer-facing portals, improving usability and transaction visibility for business users.
- Designed SQL queries and database optimization strategies using MySQL and MongoDB, to improve payment reconciliation and reporting operations, reducing query execution times by 30% for finance teams.
- Integrated third-party banking, UPI, and payment gateway APIs, to automate payment authorization and settlement workflows, improving transaction success rates and operational efficiency.
- Developed Python-based automation utilities and monitoring scripts, to streamline operational reporting and transaction analysis activities, enabling faster issue identification and improved support responsiveness.

Education

Northern Arizona University — M.S. in Computer Science | GPA: 4.0/4.0

Jan 2024 – Dec 2025

Relevant Coursework: Data Structures & Algorithms, Cloud Computing, High-Performance Computing (HPC), Blockchain